

CinePile — Full Feature Inventory

Audience: business partner (internal). Shows **every** feature in the platform — built, partial, planned, and intentionally out-of-scope. No filtering, no marketing copy. Engineering's honest snapshot.

Last updated: 2026-06-04 **Repo:** <https://github.com/mathankj/cinepile-ott> (private) **Code state:** backend 144/144 tests, e2e 108/108 tests, deployed locally for QA.

Status legend

-  **Live** — running in the code today, passes tests
 -  **Partial** — code exists but needs polish or wiring
 -  **Planned** — designed, not yet implemented
 -  **Out of scope** — deliberately not building (see "Out of scope" section)
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1. Authentication & accounts

#	Feature	Status	Notes
1.1	Email + password signup	✓	Floating-label form
1.2	Email + password sign-in	✓	JWT access + refresh tokens
1.3	Logout (single device)	✓	Clears tokens + profile
1.4	RFC-6761 reserved-TLD friendly error mapping	✓	Generic <code>.test/.example</code> rejected cleanly
1.5	Axios auto-refresh on 401	✓	Silent re-auth, retries request once
1.6	Password reset via email	📄	Needs Mailgun (free tier OK)
1.7	Email verification	📄	Needs Mailgun
1.8	Google social login	📄	Needs Google Cloud OAuth credentials
1.9	Apple sign-in	📄	Needs Apple Developer membership
1.10	Profile PIN / lock	📄	Backend column + UI
1.11	Account settings (change email/password/delete account)	📄	DPDPA compliance
1.12	Cross-device session list / "log out everywhere"	📄	Refresh-token table query
1.13	Two-factor (2FA)	✗	Out of V1; security overhead vs OTT user benefit is low
1.14	Session token rotation on refresh	✓	Backend rotates refresh tokens automatically

2. Profiles ("Who's watching?")

#	Feature	Status	Notes
2.1	Profile picker screen on login	✓	Netflix-style 2×2 grid
2.2	Profile auto-create on signup	✓	First profile is the user's full name
2.3	4 profiles per account hard cap	✓	Matches Netflix Standard tier
2.4	Avatar picker (emoji grid)	✓	16 options; expandable
2.5	Kid profile flag (adult/kid kind)	✓	Stored; enforcement pending
2.6	Profile edit (name, avatar, kind)	✓	Modal with form
2.7	Profile delete (non-primary only)	✓	Primary cannot be removed
2.8	Active-profile persistence	✓	Zustand + localStorage with hydration handling
2.9	Switch profile via navbar	✓	Dropdown in profile menu
2.10	Profile-scoped continue-watching + watchlist + reactions	📋	Schema migration (add <code>profile_id</code>)
2.11	Kid mode content filter (U-rated only)	📋	After 2.10
2.12	Profile PIN / lock	📋	After 2.10
2.13	Profile transfer between users	✗	Not a real OTT feature

3. Home / Discovery

#	Feature	Status	Notes
3.1	Billboard hero (backdrop + Play + Info CTAs)	✓	With deterministic gradient fallback when no art
3.2	Hero entrance animation (scale + fade)	✓	Framer Motion
3.3	Continue Watching row	✓	Cache-busts on every progress write
3.4	My List row	✓	Cache-busts on watchlist add/remove
3.5	New Releases row	✓	Backend window: last 30 days
3.6	Trending Now row	✓	view_count desc (proxy for trending)
3.7	Top in country row	✓	?country=IN query param
3.8	Recommended for You row	✓	Seeded by thumbs-up + watchlist + history
3.9	Because You Watched X row(s)	✓	Cap 2 rows; seeded by finished titles
3.10	Genre rows from user's top genres	✓	Auto-derived from history
3.11	Home page server-side cache (60s TTL)	✓	Per-user; auto-invalidated on writes
3.12	Skeleton shimmer while loading	✓	Site-wide style class
3.13	Trailer auto-play on hover (Netflix-iconic mini-trailer)	📋	Phase 2
3.14	Top 10 in India daily-refreshed row	📋	Phase 2
3.15	Watch Again row (completed titles)	📋	Phase 2
3.16	Editorial / curated collections	📋	Phase 3 (admin tool needed)
3.17	ML-based recommendation engine	📋	Phase 3

4. Browse / Search

#	Feature	Status	Notes
4.1	/browse with type / genre / sort filters	✓	URL state, deep-linkable
4.2	Type filter dropdown (All / Movies / TV Shows)	✓	Added in QA pass
4.3	Genre dropdown from /v1/home/genres	✓	5 genres seeded
4.4	Sort: Newest, Oldest, A→Z, Z→A, Most watched	✓	
4.5	Pagination (Prev/Next)	✓	Disabled at boundary
4.6	Empty-state message	✓	"No titles match. Try clearing filters."
4.7	Search with 300ms debounce	✓	
4.8	2-char minimum guard	✓	UI hint
4.9	LIKE-escape (% and _ are literal, not wildcards)	✓	SQL safety
4.10	Search-result highlighting	📋	Phase 2
4.11	Search history / recent searches	📋	Phase 3
4.12	Voice search	✗	Out of scope

5. Title detail / Series detail / Episodes

#	Feature	Status	Notes
5.1	Title detail page (movie + series)	✓	Synopsis, cast, genres, age, runtime
5.2	Play button (movies) / Play S1E1 (series)	✓	
5.3	More Info / Add to My List buttons	✓	
5.4	Watch Trailer button	✓	Backed by <code>trailer_url</code> ; placeholder seeded
5.5	Three-state reaction (thumbs up/down/double-up)	✓	
5.6	Episode list per season	✓	Per-season tabs
5.7	Episode runtime + intro markers	✓	
5.8	Hover-prefetch detail data	✓	Cards prefetch detail query on hover
5.9	Card hover overlay (Play/+/Info icon buttons)	✓	Real handlers; keyboard-accessible
5.10	Backwards-compatible title-by-slug lookup	⚠	Old <code>the-anjaneya-chronicles</code> slug aliased
5.11	User reviews / ratings	📋	Phase 3 (needs moderation)
5.12	Cast bio pages	📋	Phase 3
5.13	Episode lazy-load past S5 (long series)	📋	Phase 2
5.14	Trailer auto-play (muted, 30s) on title detail	📋	Phase 2

6. Video player

#	Feature	Status	Notes
6.1	HLS playback via hls.js	✓	Chrome/Firefox/Edge
6.2	Native HLS playback (Safari)	✓	iOS + macOS
6.3	Direct MP4 fallback	✓	When source isn't <code>.m3u8</code>
6.4	Adaptive bitrate (auto quality)	✓	hls.js default ABR
6.5	Manual quality picker (Auto / 1080p / 720p / 480p / ...)	✓	Settings gear top-right
6.6	Multi-audio track switcher	✓	Populates from manifest <code>EXT-X-MEDIA TYPE=AUDIO</code>
6.7	Multi-subtitle / CC switcher	✓	Sidecar <code>.vtt</code> + in-manifest both supported
6.8	Skip Intro overlay	✓	Episode markers <code>intro_start_sec</code> / <code>intro_end_sec</code>
6.9	Resume from last position	✓	Per-user, per-title
6.10	Progress reporting (every 10s + on pause)	✓	
6.11	Cosmetic anti-capture shield	✓	Right-click block, controlsList=nodownload, tab-hidden curtain
6.12	Skip Recap overlay	📋	Phase 2
6.13	Next-Episode auto-advance with countdown	📋	Phase 2
6.14	Manual quality cap toggle ("Use less data")	📋	Phase 2
6.15	Picture-in-picture	📋	Phase 3 (currently disabled for anti-capture)
6.16	Chromecast / Google Cast	📋	Phase 2
6.17	AirPlay	📋	Phase 3
6.18	Real DRM (Widevine + PlayReady + FairPlay)	⚠️	Client-side EME scaffolded; needs license-server contract
6.19	Downloads for offline	📋	Phase 2 (requires 6.18 first)
6.20	4K HDR playback	📋	Phase 3 (needs catalog + CDN scale)
6.21	Dolby Vision / Dolby Atmos	📋	Phase 3 (needs Dolby license)

7. Subscription & billing

#	Feature	Status	Notes
7.1	/subscribe page with plan list	✓	
7.2	Razorpay Orders mode	✓	One-time payment per billing period
7.3	Razorpay test-checkout HTML helper	✓	Dev only
7.4	Subscription states (active / pending / cancelled)	✓	
7.5	402 paywall on paid content with View Plans CTA	✓	
7.6	Free titles bypass subscription (<code>is_free</code> flag)	✓	
7.7	Cancel at period end	✓	
7.8	Webhook idempotency (no duplicate sub on retry)	✓	<code>webhook_events</code> table
7.9	Razorpay Subscriptions mode (recurring)	⚠	Code present; KYC required to enable
7.10	Tier ladder: Mobile / Standard / Premium	📋	Phase 2
7.11	Simultaneous-stream enforcement per tier	📋	Phase 2
7.12	Device limit per tier (1 / 2 / 4)	📋	Phase 2
7.13	Annual discount bump (17% → 33%)	📋	Phase 2; config change
7.14	Promo codes / discount campaigns	📋	Phase 3
7.15	Gift subscriptions	📋	Phase 3
7.16	Family plan (multi-email shared)	📋	Phase 3
7.17	AVOD (free tier with ads)	📋	Phase 3
7.18	Pay-per-view (TVOD)	✗	Out of scope

8. Multi-language / i18n

#	Feature	Status	Notes
8.1	i18next + react-i18next	✓	
8.2	Language picker in navbar (globe icon)	✓	
8.3	LocalStorage persistence	✓	Survives reload
8.4	Browser language auto-detect on first visit	✓	
8.5	English UI	✓	
8.6	Hindi UI	✓	
8.7	Tamil UI	✓	
8.8	Telugu / Kannada / Malayalam / Bengali UI	📋	Phase 2 (add JSON files)
8.9	Subtitle (.vtt) sidecar upload per language	✓	Admin endpoint
8.10	In-manifest subtitle tracks	✓	Player auto-discovers
8.11	Multi-language title metadata (per-title translations)	📋	Phase 3 (schema change)
8.12	Auto-translated subtitles via Whisper	📋	Phase 3
8.13	Auto-dubbed audio (AI)	📋	Phase 3
8.14	Multi-audio track upload	✗	Browser limitation; multi-audio must come from producer HLS

9. Admin / content management

#	Feature	Status	Notes
9.1	Admin login (role-gated)	✓	Role: admin / content_manager / user
9.2	Admin dashboard with stats	✓	
9.3	Admin titles list with filter + search	✓	Includes drafts
9.4	Create new title (movie or series)	✓	
9.5	Edit title metadata (everything except slug after create)	✓	
9.6	Publish / schedule / archive workflow	✓	
9.7	Soft-delete with restore	✓	deleted_at column
9.8	Video upload (multipart streamed to B2)	✓	1 GB cap
9.9	Magic-byte sniff (ISO BMFF / WebM / HLS validation)	✓	Defense against renamed EXE
9.10	Subtitle (.vtt) upload per language	✓	5 MB cap, upsert by language
9.11	Subtitle remove (DB row only; object retained for undo)	✓	
9.12	Audio track management UI	⚠	Backend endpoint exists; admin UI not yet exposed
9.13	Per-episode video upload	✓	
9.14	Per-episode subtitle upload	✓	
9.15	Season + episode create	✓	
9.16	Genre management	✓	
9.17	Admin-scoped detail endpoint for drafts	✓	Fixes "editor blank for new title" bug
9.18	User management (list, change role)	✓	Admin-only
9.19	Audit log (who/what/when/before/after)	✓	Filterable
9.20	Bulk operations (delete N, publish N)	📋	Phase 3
9.21	Editorial collection builder	📋	Phase 3
9.22	Content moderation / takedown workflow	📋	Phase 3
9.23	Title import from CSV / API	📋	Phase 3

10. Watch progress / history

#	Feature	Status	Notes
10.1	Movie progress save	✓	Per-user
10.2	Episode progress save	✓	
10.3	Resume from last position on next play	✓	
10.4	Continue Watching list (collapsed by series)	✓	
10.5	Completion threshold (~95%) marks as completed	✓	
10.6	"Remove from Continue Watching"	✓	
10.7	Full viewing history page	✓	Paginated
10.8	Delete from history	✓	
10.9	Export viewing history	📋	Phase 3 (DPDPA compliance)

11. Watchlist / My List

#	Feature	Status	Notes
11.1	Add to My List	✓	Idempotent
11.2	Remove from My List	✓	
11.3	My List page with shimmer + empty state	✓	Includes CTA to /browse
11.4	+ button on card flips to ✓ when on list	✓	
11.5	My List filter + sort	📋	Phase 2

12. Reactions

#	Feature	Status	Notes
12.1	Thumbs up	✓	
12.2	Thumbs down	✓	
12.3	Double thumbs up ("love")	✓	
12.4	Reaction history (/v1/me/reactions)	✓	
12.5	Reactions used as recommendation seeds	✓	Fixed in QA pass (was filtering on non-existent <code>kind="like"</code>)
12.6	Public ratings aggregate	✗	Netflix doesn't do this; we follow

13. Frontend infrastructure

#	Feature	Status	Notes
13.1	React 19 + Vite + TypeScript + Tailwind v4	✓	
13.2	React Router 7 with lazy-loaded routes	✓	Per-route code splitting
13.3	TanStack Query for server state	✓	
13.4	Zustand for auth + profile state	✓	With persist + hydration handling
13.5	Framer Motion animations	✓	Billboard entrance, row stagger, card hover
13.6	Responsive design (5 viewports tested)	✓	360 / 414 / 768 / 1280 / 1920
13.7	Mobile + tablet drawer (hamburger)	✓	Slides from right
13.8	Skeleton shimmer site-wide	⚠	Home + MyList have it; other pages still "Loading..."
13.9	Error toasts on API failures	📋	Currently silent on /admin/users 500
13.10	Dark mode (default, OTT-style)	✓	
13.11	Light mode toggle	✗	Out of scope for OTT
13.12	Service worker / PWA	📋	Phase 3

14. Backend infrastructure

#	Feature	Status	Notes
14.1	FastAPI async	✓	
14.2	SQLAlchemy 2.0 async ORM	✓	
14.3	Postgres (Neon for dev)	✓	
14.4	Alembic migrations	✓	
14.5	Backblaze B2 storage with presigned URLs	✓	
14.6	S3-compatible storage abstraction (R2, AWS, Storj)	✓	Works with any S3 endpoint
14.7	JWT auth with rotation	✓	
14.8	Audit logging service	✓	Every admin action logged
14.9	Webhook idempotency	✓	<code>webhook_events</code> table with seen-id check
14.10	Health + readyz endpoints	✓	
14.11	Razorpay client (Orders + Subscriptions modes)	✓	Subscriptions mode pending KYC
14.12	Rate limiting	📋	Phase 2 (Redis-keyed)
14.13	Background job queue (Celery / RQ / Arq)	📋	Phase 3
14.14	Email service abstraction	📋	Phase 2 (Mailgun)
14.15	DRM token signing service	✓	Code path; activated when provider configured
14.16	Search service (Postgres full-text)	✓	
14.17	Elasticsearch / Meilisearch (better search)	📋	Phase 3
14.18	CORS configured	✓	
14.19	Request ID logging	✓	
14.20	Structured logging (JSON)	✓	structlog

15. Testing & quality

#	Feature	Status	Notes
15.1	Backend pytest suite (integration + unit)	✓	144/144 passing
15.2	Backend load tests (Locust)	⚠	Baseline recorded; not in CI
15.3	Frontend Playwright e2e	✓	108/108 desktop, 31/95 mobile (skips are by design)
15.4	Role × route access matrix tests	✓	56 access checks
15.5	Visual responsive audit screenshots	✓	All 5 viewports captured
15.6	Real upload e2e (B2 round-trip + presigned URL verify)	✓	
15.7	Subtitle upload e2e (4 tests)	✓	
15.8	i18n switch + persist e2e	✓	
15.9	CI/CD pipeline (GitHub Actions)	📄	Phase 2
15.10	Production smoke tests	📄	After Phase 1 deploy
15.11	Performance budget tests (LCP, CLS, INP)	📄	Phase 3
15.12	A11y automated tests (axe-core)	📄	Phase 2

16. Infrastructure & ops

#	Feature	Status	Notes
16.1	Local dev Docker Compose	✗	Not provided; npm + python venv only
16.2	Production deployment runbook	⚠	Documented; not yet executed
16.3	Database backup strategy	📋	Phase 2 (Neon has point-in-time but documented)
16.4	Multi-region CDN (Bunny / Cloudflare)	📋	Phase 2 (single-region works for V1)
16.5	Transcoding pipeline (MediaConvert)	📋	Phase 3
16.6	Status page (statuspage.io / Statuspage)	📋	Phase 3
16.7	Monitoring (Sentry)	📋	Phase 2 (free tier OK)
16.8	Product analytics (PostHog)	📋	Phase 2
16.9	Uptime monitoring (Better Uptime / Pingdom)	📋	Phase 2
16.10	Blue-green deployment	📋	Phase 3
16.11	Auto-scaling	📋	Phase 3 (Contabo vertical-scale fine for now)
16.12	Database replicas	📋	Phase 3
16.13	Forensic watermark (invisible)	📋	Phase 3 (₹15k+/mo)
16.14	Anti-piracy monitoring	📋	Phase 3

17. Native apps & TV

#	Feature	Status	Notes
17.1	iOS native app	📋	Phase 3 (8-12 weeks)
17.2	Android native app	📋	Phase 3 (8-12 weeks)
17.3	Samsung Tizen TV app	📋	Phase 3
17.4	LG webOS TV app	📋	Phase 3
17.5	Android TV / Google TV	📋	Phase 3
17.6	Apple TV app	📋	Phase 3
17.7	Roku app	📋	Phase 3 (India market: small)
17.8	Amazon Fire TV	📋	Phase 3

18. Social / community

#	Feature	Status	Notes
18.1	Reactions (thumbs)	✓	Per-user, not public
18.2	Group Watch / Watch Party	📋	Phase 3
18.3	Activity sharing ("I just watched X")	📋	Phase 3 (opt-in)
18.4	User reviews + ratings	📋	Phase 3 (needs moderation)
18.5	Follow other users	✗	Not an OTT feature
18.6	Comments per title	✗	Toxic without heavy moderation

19. Legal / compliance

#	Feature	Status	Notes
19.1	GDPR / DPDPA data export	📋	Phase 3
19.2	Account deletion with data removal	📋	Phase 3
19.3	Cookie consent banner	📋	Phase 2 (EU users, even one)
19.4	Terms of Service page	📋	Phase 1 (footer link exists; page empty)
19.5	Privacy Policy page	📋	Phase 1
19.6	Cookie Policy page	📋	Phase 1
19.7	Content rating per region	📋	Phase 3 (CBFC + age-rating per market)
19.8	Geographic content restrictions	📋	Phase 3 (CDN geo-block)

Out of scope (we will NOT build, even if asked)

Feature	Reason
Live sports / IPL	Hotstar's moat — rights ₹400+ cr/yr
Voice search	Indian-accent English voice recognition is rough
Smart-speaker integration	Niche for video
Cloud DVR for live	Premature; no live yet
Pay-per-view (TVOD)	Doesn't fit SVOD model
Two-factor auth (2FA)	Overhead vs benefit low for OTT
Light mode toggle	OTT is dark by default
Real-time comments per title	Moderation cost too high
Follow other users	Not a streaming feature

Summary table

Status	Count
✓ Live	104 features
⚠ Partial	6 features
📅 Planned	~85 features
✗ Out of scope	9 features
Total inventoried	~204 features

Where to find what

Doc	Purpose
docs/business/partner-full-features.md	This file (full inventory for partner)
docs/business/client-pitch.md	Phased pitch deck for client meeting
docs/qa/FINAL-REPORT.md	QA findings + cost decisions for client
docs/qa/competitive-analysis.md	Indian OTT market analysis
docs/decisions/0003-drm.md	DRM provider comparison + ADR
docs/api/v1.md	API contract
docs/db-schema.md	Database schema reference

Repo: <https://github.com/mathankj/cinepile-ott> (private)